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Full life-cycle cutting force prediction in ball helical milling based on oblique cutting analysis

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Research Article

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Full life-cycle cutting force prediction in ball helical milling based on oblique cutting analysis

Zhitong Zhou, Haiyan Wang*

Abstract

In order to predict the change of cutting force in ball helical milling of titanium alloy, considering the kinematics of helical milling and the geometric model of ball end cutter, the cutting process is divided into three stages, cutting-in, steady and cutting-out stages, and the shape and volume of undeformed chip in different stages are described. Then according to the oblique cutting theory and the principle of maximum shear stress, the constraint relationships among the parameters such as friction angle, shear angle and shear stress in the micro-element are established. Thus based on the position parameters at different stages, a three-direction full life-cycle cutting force model is established, the cutting force coefficients are identified and expressed as Weibull functions of instantaneous undeformed chip thickness. Finally, the predicted cutting forces are compared with the experimental values. The results show that the maximum average error is less than 18%, which verify the accuracy of the model.

Keywords Full life-cycle cutting force model; Ball helical milling; Oblique cutting; Kinematics model; Material removal mechanism

Nomenclature

n_{rot}(rpm)	Rotation speed
n_{orb}(rpm)	Orbital revolution speed
v_t, v_a(m/min)	Tangential and axial feed speed
f_t, f_a(mm/t)	Tangential and axial feed per tooth
R_h, R_t(mm)	Radius of hole and tool
N	Tooth number of the tool
γ(radian)	Helix angle of the tool trajectory
a_o(mm/orb)	Axial depth per orbital revolution
i (radian)	Helix angle of the tool
z(mm)	Axial height of point P
z_1, z_2(mm)	Axial depths of cut at time t and t-t_{orb}
$R_s(z)$(mm)	Radial distance of point P
$\kappa(z)$(radian)	Axial immersion angle of point P
$\psi(z)$(radian)	Radial lag angle of point P
$\xi_j(z)$, $\theta_j(z)$(radian)	Radial position angle of point P in o-xyz and a-$x_a y_a z_a$
t(s)	Tool movement time
e(mm)	Eccentricity
ζ(radian)	Tool orbital revolution angle
ζ_0(radian)	Initial angle of tool orbital revolution angle
F_X, F_Y, F_Z(N)	Cutting forces in the x,y,z directions
H_t(mm)	Workpiece thickness
dS(mm)	Discrete cutting arc length
db(mm)	Projection length of the micro-element along the cutting speed direction
h_j(mm)	Instantaneous undeformed chip thickness
$F_d(t)$(N)	Indentation force
ΔS(mm^2)	Cross-section area of the material to be cut
K_{db}(N/mm^2)	Indentation force coefficient
K_{tc}, K_{rc}, K_{ac}(N/mm^2)	Shear force coefficients of tangential, radial and axial forces

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K_{te}, K_{re}, K_{ae} (N/mm) Edge force coefficients of tangential, radial and axial forces
α, α_n (radian) Rake angle and normal rake angle
β_i, β_n (radian) Friction angle and normal friction angle
φ_i, φ_n (radian) Shear angle and normal shear angle
τ(MPa) Shear stress
S_S (mm^2) Shear area
η(radian) Chip flow angle
r_e (μm) Edge radius of the tool
$dF_{j,z}, dF_{s,j,z}, dF_{f,j,z}, dF_{n,j,z}$ (N) Resultant, shear, friction, and normal micro forces applied at the chip by the flute j_{th} at an axial depth of z
$dF_{t,j,z}, dF_{r,j,z}, dF_{a,j,z}$ (N) Tangential, radial and axial micro forces for the flute j at an axial depth of z
$dF_{tn,j,z}, dF_{rn,j,z}, dF_{an,j,z}$ (N) Projections of $dF_{t,j,z}, dF_{r,j,z}$, and $dF_{a,j,z}$ on normal plane
A_1, A_2, A_3, T Transformation matrixs
ϕ_i (radian) Angle between $dF_{j,z}$ and the normal plane
ϕ_n (radian) Angle between the projection of $dF_{j,z}$ on the normal plane and the cutting plane

1 Introduction

Titanium alloy is widely used in aerospace field due to its excellent material properties such as high specific strength, high specific stiffness, corrosion resistance and heat resistance [1]. However, Ti-6Al-4V is a typical difficult-to-machine material with low thermal conductivity, high chemical activity at high temperature and small elastic modulus. Helical milling is a new hole-making technology for difficult-to-cut material emerged in recent years，it has the advantages of reducing the axial force, improving chip removal and heat dissipation conditions [2].

Brinksmeier et al. [3] utilized mathematical theory to analyze the kinematics of helical milling, calculated the ratio of the material volume removed by the bottom and the side edges of the tool, which is only related to hole and tool diameters. Wang et al. [4] established the cutting force model in the helical milling process for the first time, applied the average force method to identify cutting force coefficients, assured the correctness of the model through cutting experiments. Rey et al. [5] established a nonlinear cutting force model, compared the predicted cutting force with the experimental values. It was noted that the average errors of radial force and axial force were 18.8% and 5.3%, respectively. Liu et al. [6] proposed the time-domain cutting force model of helical milling, checked the accuracy of the model through experiments. Zhou et al. [7] divided helical milling into three stages based on the dedicated cutting tool, reckoned the undeformed chip thickness and volume of each stage, discussed the relationships with cutting force and hole-making quality, and verified its accuracy through experiments. Ji et al. [8] described a simulation method for different cutting stages with same failure parameters, it was presented that the difference between the predicted and the experimented values were less than 10%.

In the prediction of cutting force, the solution of cutting force coefficient is very important [9]. Budak and Altinta [10] put forward the average cutting force method and the oblique cutting method to solve the cutting force coefficients. Wang et al. [11] used the average force method to inverse the cutting force coefficients according to a series of slot-milling experiments. Adem et al. [12] proposed linear and nonlinear force models, utilized the average force method and optimization method to determine the cutting force coefficients. Lee and Altinta [13] used the oblique cutting method to calculate the cutting force coefficients of ball end cutter, then forecasted the cutting force, verified the correctness through more than 60 tests. Srinivasa and Shunmugam [14] based on the material property of low carbon steel and the principle of oblique cutting, adopted Newton–Raphson algorithm, minimum energy principle, and JC model to determine friction angle, shear angle and shear stress, constructed a cutting force model for micro end-milling. Luo et al. [15] carried out three sets of

linear end milling experiments, calibrated the shear stress, shear angle and friction angle as the double exponential functions of the instantaneous undeformed chip thickness, solved the cutting force by utilizing the oblique cutting coefficient identification formulas.

In the ball helical milling process, the ball end cutter takes longer time to cutting-in and out of the workpiece, then the cutting forces change more slowly than the end cutter. In addition, the change of cutting forces have a great impact on the entry and exit of the hole-making quality. Therefore, it is very important to predict the cutting force in ball helical milling process. David et al. [16] analyzed cutting force, hole accuracy, hole wall surface quality, exit burr and tool life in the ball helical milling process, the results showed that ball helical milling was a feasible choice. Wang et al. [17] investigated ball helical milling process and set up cutting force model, but the research aimed at steady stage in the cutting process, related studies was not enough.

In this paper, the ball helical milling process will be divided into three stages according to the machining time. Combined with the kinematics of the helical milling and the geometric model of the ball end cutter, the full life-cycle cutting force model in the ball helical milling of Ti-6Al-4V will be constructed, and the oblique cutting theory is utilized to calibrate the cutting force coefficients.

2.Kinematics of ball helical milling

2.1 Helical milling mechanism

Ball helical milling process is illustrated in Fig.1a. It can be seen that the tool revolves around the center of the hole while rotating, maintaining the axial feed, that is, the tool orbits along the helical trajectory while rotating, so that the hole-making diameter can be conveniently adjusted by changing the eccentricity of the tool.

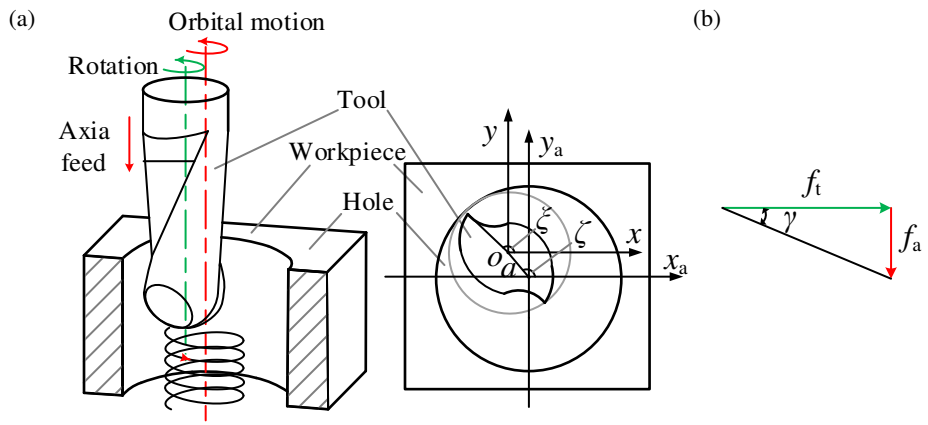

Fig. 1 Ball helical milling process.

Tangential feed per tooth f_t and axial feed per tooth f_a can be calculated as follows

$$\begin{cases} f_t = \dfrac{v_t}{n_{rot}N} = \dfrac{2\pi(R_h - R_t)}{n_{rot}N} \\ f_a = \dfrac{v_a}{n_{rot}N} = \dfrac{a_o \cdot n_{orb}}{n_{rot}N} \end{cases} \tag{1}$$

The helix angle of the tool trajectory is shown in Fig.1b, which can be expressed as

$$\gamma = \arctan\frac{a_o}{2\pi(R_h - R_t)} = \arctan\frac{v_a}{v_t} \tag{2}$$

2.2 Geometric model of the ball end cutter

As shown in Fig. 2, the geometric contour of the ball end cutter can be divided into two segments along the axial direction, namely NS and OS, o is the tool tip.

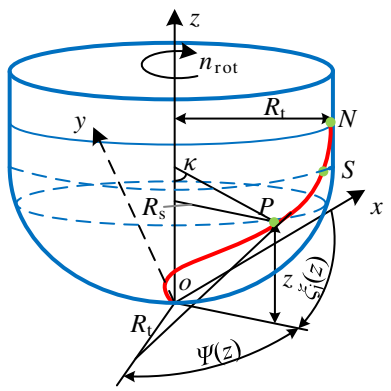

Fig. 2 Geometry of ball end cutter.

For any point P on the j_{th} cutting edge, its position is determined by axial distance z, radial distance $R_s(z)$, axial immersion angle $\kappa(z)$, radial position angle $\xi_j(z)$, which can be obtained as

$$\begin{cases} \psi(z) = z \cdot \tan i / R_t \\ \xi_j(z) = \xi_0 + \dfrac{2\pi \cdot n_{rot} t}{60} + \dfrac{(j-1)\cdot 2\pi}{N} - \psi(z) \\ R_s(z) = \begin{cases} R_t\sqrt{1-(1-z/R_t)^2}, 0 \le z < R_t \\ R_t, z \ge R_t \end{cases} \\ \kappa(z) = \begin{cases} \sin^{-1}\dfrac{R_s(z)}{R_t}, 0 \le z < R_t \\ \pi/2, z \ge R_t \end{cases} \end{cases} \quad (3)$$

2.3 Motion equation of cutting edge

In order to clearly analyze the cutting process, two coordinate systems are established, the hole center is the coordinate origin to establish the workpiece coordinates $a\text{-}x_a y_a z_a$, as shown in Fig. 1. The tool coordinate system $o\text{-}xyz$ is described with the tool tip point of the ball end cutter as the coordinate origin, and the x, y, and z axes are parallel to the workpiece coordinate system. According to the kinematics of helical milling and the geometric model of the ball end cutter, the motion trajectory equation of any point P can be defined as

$$\begin{cases} x_p = R_t \cdot \sin(\kappa) \cdot \cos\xi_j(z) + e \cdot \cos\zeta \\ y_p = R_t \cdot \sin(\kappa) \cdot \sin\xi_j(z) + e \cdot \sin\zeta \\ z_p = z \end{cases} \quad (4)$$

then

$$\zeta(t) = \zeta_0 + 2\pi n_{orb} t / 60 \quad (5)$$

The motion trajectories of the point S and the tool tip o in Fig. 2 are illustrated in Fig. 3. It can be seen that the cutting depth of the ball end cutter is the sum of the workpiece length and the radius of the ball end cutter. Fig.3a is the path of the entire machining process, Fig.3b is the track of two points in two orbital revolution periods. It can be concluded that the path of the point o is equal to the helical trajectory of the tool.

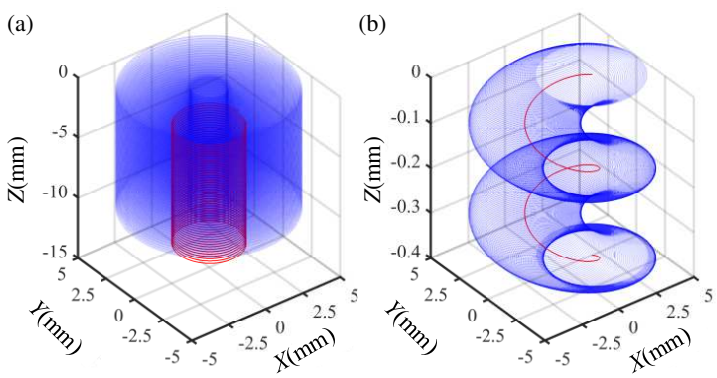

Fig. 3 Motion simulation of ball helical milling

2.4 Distribution of chip thickness

Through kinematics analysis, it can be presented that the S-shaped cutting edge of ball end cutter participates in cutting, and the cutting process is similar to inclined milling. A dynamic geometric model of chip thickness is established.

An expanded view of the tool path in ball helical milling for one orbital revolution period is shown in

Fig.4a. It can be seen that during the whole machining process, the spherical section of the tool removes tangential and axial materials at feed rates v_t and v_a simultaneously, different feed directions will produce diverse chip thicknesses. The chip thickness due to tangential feed is shown in Fig.4b, it can be discribed as

$$h_{cxj} = f_t \cdot \xi_j \cdot \sin(\xi_j)\sin(\kappa) / 2\pi n_{rot} \quad (6)$$

Chip thickness caused by axial feed is shown in Fig.4c, the tool from one position to another is described in detail, while the chip thickness does not alter with the instantaneous immersion angle due to vertical feed. Chip thickness can be calculated as follows

$$h_{czj} = f_a \cdot \xi_j \cdot \cos(\kappa) / 2\pi n_{rot} \quad (7)$$

As shown in Fig.4d, the total chip thickness can be obtained by adding the chip thicknesses produced by tangential and axial motions

$$h_j = \frac{f_t \cdot \xi_j \cdot \sin(\xi_j)\sin(\kappa) + f_a \cdot \xi_j \cdot \cos(\kappa)}{2\pi n_{rot}} \quad (8)$$

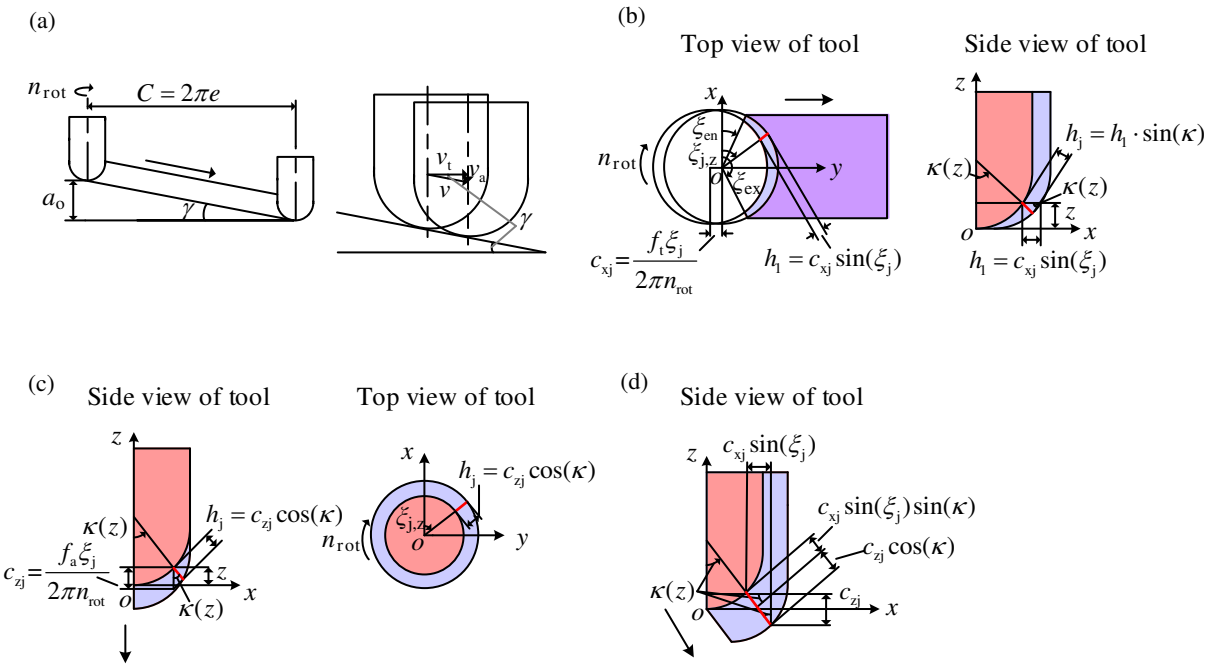

Fig. 4 Chip thickness in different directions. **a** Expanded diagram of an orbital revolution trajectory. **b** Chip thickness due to tangential feed. **c** Axial feed. **d** Combined tangential and axial feeds.

3. Experiment of ball helical milling

3.1 Experimental procedures

The experiments were carried out on a five-axis machining center (DMU80T), the experiment workpiece material was Ti-6Al-4V with 120mm×120mm×5mm, the two-fluted ball end cutter with 35° helix angle and 6 mm diameter was utilized, the cutting parameters are illustrated in Table 1.

Table 1 Experiment parameters

Factors	Levels		
	1	2	3
Rotation speed n_{rot} (rpm)	3183	3714	4244
Feed rate per tooth f_t (mm/t)	0.05	0.06	0.08
Axial depth per orbital revolution a_o (mm/orb)	0.1	0.15	0.2

The cutting forces were measured during the helical milling process, which were monitored and collected in real time by a Kistler-9257B dynamometer, the experiment setup is shown in Fig. 5.

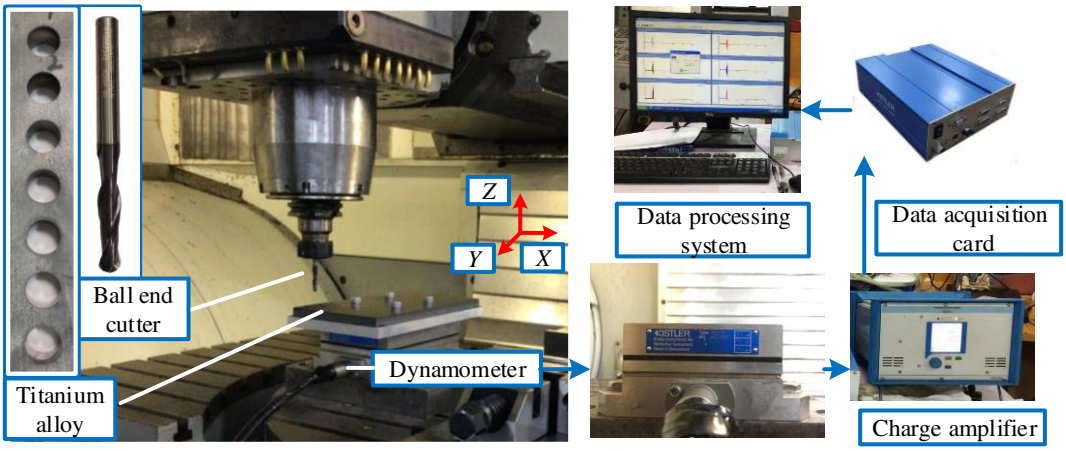

Fig. 5 Experimental setup

3.2 Analysis of experiment results

3.2.1 Cutting forces

When the rotating speed is 4244 rpm, the feed per tooth is 0.06 mm/t, and the cutting depth is 0.2 mm/orb, the cutting forces are shown in Fig. 6. It can be seen that the fluctuation period of the three-direction forces is equal to the time of one orbital revolution of the tool, and the total machining time of a single hole is 62.1 s, in other words, the total number of orbital revolutions is 44.69 cycles. At the 16.76th orbital revolution, the ball end cutter completely cuts into the workpiece, the amplitude of F_X, F_Y increases from 0 N to 20 N, the average value of F_Z increases from 0 N to 320 N, the amplitude increase from 0 N to 120 N. In the process of 23.29-38.81 s, the forces keep steady, the maximum values of F_X, F_Y are 20 N, the minimum values are -20 N, and the maximum values of F_Z are 320 N, the minimum values are 200 N. In the course of 38.81-62.1s, the forces fluctuation is just opposite to the first 16.76 orbital revolution cycles and the S-shaped edge cut out the workpiece slowly, and the forces gradually decrease to 0 N, so when utilizing ball end cutter, the geometric error at the entry and exit will be improved [18].

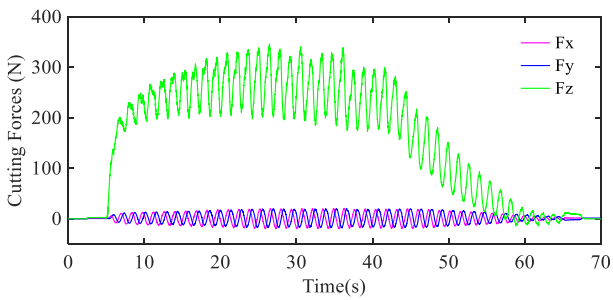

Fig. 6 Cutting forces in ball helical milling

3.2.2 Tool wear

As a difficult-to-cut material, titanium alloy has a small cutting deformation coefficient and a small thermal conductivity coefficient, which will result in the rapid wear of the tool in the cutting process. Adhesion, oxidation and abrasive wear are main wear forms. In addition, adhesion and abrasive wear will lead to micro collapse blade, even to tool failure [19]. It can be seen from Fig.7a that a large amount of titanium alloy powder was bonded to the cutter edge. During the machining process, the tool and the workpiece rubbed violently, the temperature and pressure in the cutting area would be high, then the surface metal near the cutter edge would be softened, thus the material would stick to the cutter edge, and then fall off while taking away part of the tool material, resulting in adhesion wear. In addition, there were some scratches on the tool surface, the main reason

lied in the tool and Ti-6Al-4V moved relatively quickly in the cutting process, the hard spots and adhesion material left scratches on the tool surface, leading to abrasive wear. As shown in Fig.7b, a small piece of material fell off from the cutting edge, resulting in micro collapse blade, but the overall structure of the tool was complete and could be processed normally.

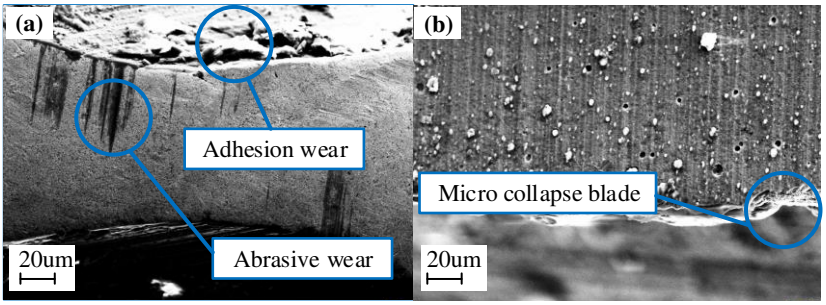

Fig. 7 Tool wear

3.2.3 Quality of hole-making

The inner micrometer is used to measure and record the hole size change of titanium alloy. The entry and exit diameters of 27 holes are selected as measurement objects, the detection results are shown in Fig. 8.

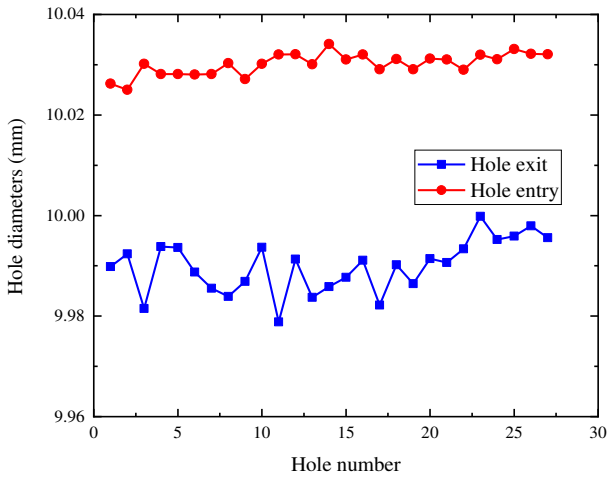

Fig. 8 Entry and exit diameters of holes

It can be seen that the maximum of entry diameter is 10.034 mm and the minimum is 10.025 mm; maximum of exit diameter is 9.999 mm and minimum is 9.979 mm. The average entry diameter of the hole is larger than the exit diameter as a whole, with the change of helical milling depth, the cutter is subjected to gradually changing radial force, and then the degree of flexural deformation is changed constantly, resulting in the actual hole-making shape of tapered hole. The experimental results show that the tolerance levels are IT6-IT8, which are equivalent to the accuracy level of hole-making and meet the machining requirements.

4 Full life-cycle cutting force modeling of ball helical milling

4.1 Three stages during ball helical milling

When the rotating speed is 3183 rpm, the feed per tooth is 0.08 mm/t, the cutting depth is 0.2 mm/orb, the three-direction cutting forces and the positions of the tool in the cutting process are shown in Fig. 9.

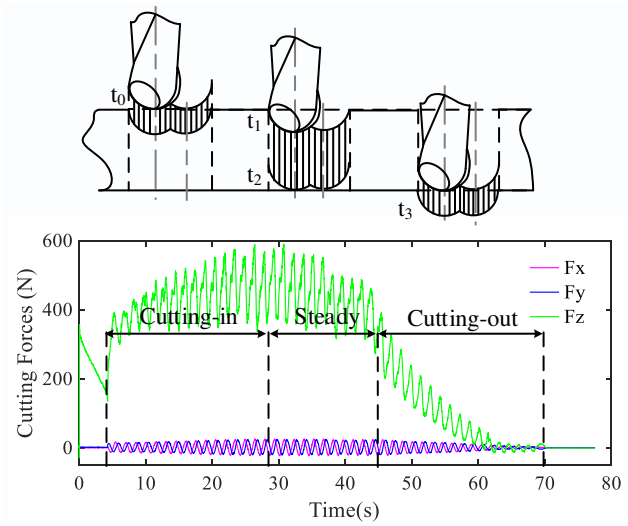

Fig. 9 Three machining stages in ball helical milling process

The maximum axial force is about 600 N, the cutting time is from 4.3 s to 70.1 s, the cutting force in any direction shows a changing trend from small to large, gradually stable and from large to small, especially the axial force. Therefore, the whole cutting process can be divided into three stages, cutting-in, steady and cutting-out stages, the corresponding cutting positions of the tool are also shown in Fig. 9.

Taking the tool tip contacting the Ti-6Al-4V surface as starting point, the machining time of each stage and the entire machining times can be written as

$$\begin{cases} \Delta T_1 = t_1 - t_0 = R_t / f_a \\ \Delta T_2 = t_2 - t_1 = (H_t - R_t) / f_a \\ \Delta T_3 = t_3 - t_2 = R_t / f_a \\ \Delta T = \Delta T_1 + \Delta T_2 + \Delta T_3 = t_3 - t_0 = (H_t + R_t) / f_a \end{cases} \tag{9}$$

4.2 Undeformed cutting area at three stages

Due to the irregular shape of the undeformed chip, the volume is obtained by calculating the area multiplying by the tangential feed per tooth. It is divided into three stages, as shown in Fig. 10, the coordinate equations of the two circles are

$$\begin{cases} o_1 : (x_1 - R_t)^2 + (y_1 - R_t - H_t + v_a \cdot t)^2 = (R_t)^2 \\ o_2 : (x_2 - R_t)^2 + (y_2 - R_t - H_t + z_2)^2 = (R_t)^2 \end{cases} \tag{10}$$

with

$$z_2 = z_1 - a_o = v_a \cdot t - a_o \tag{11}$$

In the cutting-in stage, the cutting depth is less than R_t, the max radius of the ball head part has not yet cut into the workpiece, and the maximum diameter of the hole-making is also not reached. The intersection points between the tool and the workpiece at time t are A and D, B and C are the contact points between the cutting edge and the upper surface of the workpiece at time $t - t_{orb}$. Meanwhile, ΔS_{ABCD} represents the cross-sectional area of the material to be cut, the undeformed chip area and the volume of the removed material increase slowly.

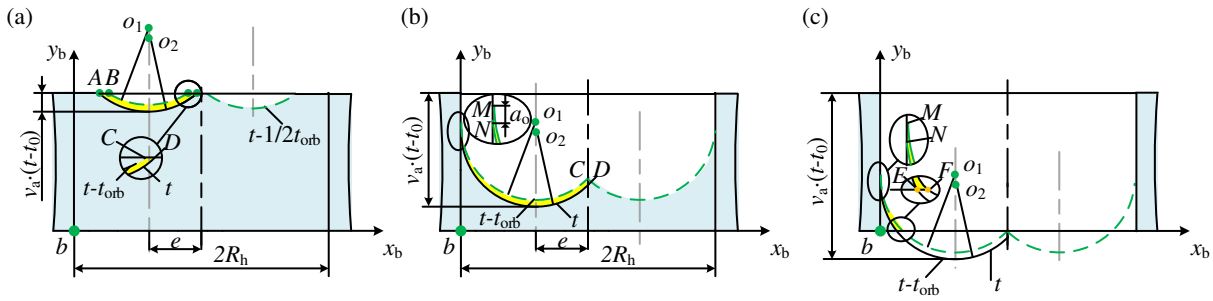

Fig. 10 Undeformed chips area at three locations

In Fig.10(b), the tool goes into the steady stage, the cutting depth is greater than the tool radius and less than the workpiece thickness, the intersection points of the tool with the hole wall at time t and $t - t_{orb}$ are N and M, the contact points with the center line of the hole are C and D, at this moment, the ΔS_{MNCD} represents the cross-sectional area of the material to be cut, during this stage, the undeformed chip area reaches to the maximum value, the removed material volume is the largest and remains unchanged. In Fig.10(c), the cutting depth is greater than the workpiece thickness H_t and less than the total cutting depth $H_t + R_t$, the intersection of the tool with the hole wall at time t and $t - t_{orb}$ are points N and M, the contact points with the lower surface of the workpiece are points E and F, and ΔS_{MNEF} represents the cross-sectional area of the material to be cut.

The undeformed cutting areas of the three analyzed positions can be determined as

$$\begin{cases} \Delta S_{\mathrm{ABCD}} = \int_{R_{\mathrm{h}}-e-R_{\mathrm{s}}(z_1)}^{R_{\mathrm{h}}-e+R_{\mathrm{s}}(z_1)} (H_{\mathrm{t}} - y_1)\mathrm{d}x - \int_{R_{\mathrm{h}}-e-R_{\mathrm{s}}(z_2)}^{R_{\mathrm{h}}-e+R_{\mathrm{s}}(z_2)} (H_{\mathrm{t}} - y_2)\mathrm{d}x & a_{\mathrm{o}} / v_{\mathrm{a}} < t < (R_{\mathrm{t}} - \sqrt{R_{\mathrm{t}}^2 - e^2}) / v_{\mathrm{a}} \\ \Delta S_{\mathrm{MNCD}} = a_{\mathrm{o}} \cdot (R_{\mathrm{t}} + e) & R_{\mathrm{t}} / v_{\mathrm{a}} < t < H_{\mathrm{t}} / v_{\mathrm{a}} \\ \Delta S_{\mathrm{MNEF}} = a_{\mathrm{o}} \cdot (R_{\mathrm{t}} + e) - \left[\int_{R_{\mathrm{h}}-e-R_{\mathrm{s}}(z_1)}^{R_{\mathrm{h}}} (H_{\mathrm{t}} - y_1)\mathrm{d}x - \int_{R_{\mathrm{h}}-e-R_{\mathrm{s}}(z_2)}^{R_{\mathrm{h}}} (H_{\mathrm{t}} - y_2)\mathrm{d}x) \right]_{t=t-H_{\mathrm{t}}/v_{\mathrm{a}}} & \\ & H_{\mathrm{t}} / v_{\mathrm{a}} + (R_{\mathrm{t}} - \sqrt{R_{\mathrm{t}}^2 - e^2}) < t < (H_{\mathrm{t}} + R_{\mathrm{t}}) / v_{\mathrm{a}} \end{cases} \tag{12}$$

4.3 Position parameters at three stages

Helical milling belongs to intermittent cutting, the contact state of each micro-element is time-varying in the cutting process. Therefore, the parameters such as axial distance, radial distance, axial immersion angle, radial position angle, width, arc length variation in real time with the change of cutting state.

4.3.1 Cutting-in stage (t_0-t_1)

Cutting-in stage begins with the tip touching the upper surface of Ti-6Al-4V and ends after the moment that the radial distance reaches the maximum. The working length of S-shaped edge increases gradually with time, that is, the number of elements involved in cutting increases gradually. The range of axial height, axial immersion angle, radial distance of micro-element at this stage are

$$\begin{cases} 0 \le z \le v_{\mathrm{a}} \cdot t \\ 0 \le \kappa \le \sin^{-1}(\sqrt{1-(1-v_{\mathrm{a}} \cdot t / R_{\mathrm{t}})^2}) \\ 0 \le R_{\mathrm{s}} \le \sqrt{v_{\mathrm{a}} \cdot t(2R_{\mathrm{t}} - v_{\mathrm{a}} \cdot t)} \end{cases} \tag{13}$$

4.3.2 Steady stage (t_1-t_2)

Steady stage starts from the moment that the radial distance reaches the maximum, ends when the tip contacts the lower surface of Ti-6Al-4V. The S-shaped edges and part of the circumferential edges cut into the workpiece vertically, the effect of the circumferential edges are to trim the hole wall, while the S-shaped edges cut the material, so the position parameters of the micro-element involved in cutting are as blow

$$\begin{cases} 0 \le z \le R_{\mathrm{t}} \\ 0 \le \kappa \le \pi / 2 \\ 0 \le R_{\mathrm{s}} \le R_{\mathrm{t}} \end{cases} \tag{14}$$

4.3.3 Cutting-out stage (t_2-t_3)

Cutting-out stage begins at the moment when the tool tip is cut out from the Ti-6Al-4V, ends at the occasion when all the S-shaped edges cut out the workpiece. The working length of S-shaped edges decrease gradually with time, in other words, it starts to cut out the workpiece from the tip. The cutting states of micro-element in this stage are just opposite to those at the cutting-in stage, then

$$\begin{cases} v_{\mathrm{a}} \cdot t - H_{\mathrm{t}} \le z \le R_{\mathrm{t}} \\ \sin^{-1}(\sqrt{1-(1-v_{\mathrm{a}} \cdot t / R_{\mathrm{t}})^2}) \le \kappa \le \pi / 2 \\ \sqrt{v_{\mathrm{a}} \cdot t(2R_{\mathrm{t}} - v_{\mathrm{a}} \cdot t)} \le R_{\mathrm{s}} \le R_{\mathrm{t}} \end{cases} \tag{15}$$

4.4 Prediction of cutting forces

To obtain the total cutting force on the j_{th} cutting edge of the ball end cutter, the tool is discretized into several micro-element distributed along the axial direction, as shown in Fig. 11. The micro-element involved in cutting are determined by the contact area between the tool envelope and the workpiece surface.

Assuming that point P is located on the j_{th} cutting edge and its axial height is z, then the cutting forces of tangential, radial and axial micro-element are established respectively

$$\begin{cases} \mathrm{d}F_{\mathrm{t,j}}(\xi, z, \kappa) = K_{\mathrm{tc}} h_{\mathrm{j}}(\xi, z, \kappa)\mathrm{d}b + K_{\mathrm{te}}\mathrm{d}S \\ \mathrm{d}F_{\mathrm{r,j}}(\xi, z, \kappa) = K_{\mathrm{rc}} h_{\mathrm{j}}(\xi, z, \kappa)\mathrm{d}b + K_{\mathrm{re}}\mathrm{d}S \\ \mathrm{d}F_{\mathrm{a,j}}(\xi, z, \kappa) = K_{\mathrm{ac}} h_{\mathrm{j}}(\xi, z, \kappa)\mathrm{d}b + K_{\mathrm{ae}}\mathrm{d}S \end{cases} \tag{16}$$

where

$$\mathrm{d}b = \mathrm{d}z / \sin\kappa \tag{17}$$

$$\mathrm{d}S(z) = \mathrm{d}z\sqrt{(R(z)\psi'(z))^2 + (R'(z))^2 + 1} \tag{18}$$

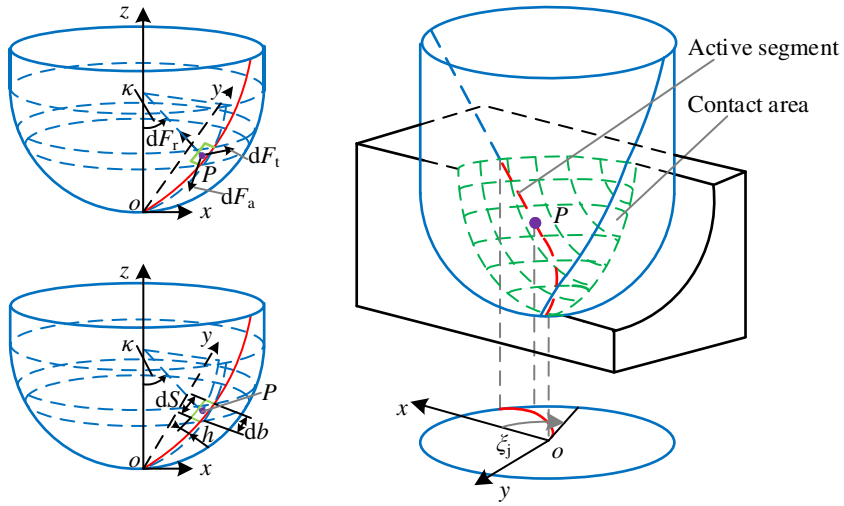

Fig. 11 Cutting edge segment and contact area

After coordinate transformation of micro-element force on the cutting edge, Eq. (16) can be transformed as follows

$$\begin{bmatrix} \mathrm{d}F_{\mathrm{xb,j}}(\xi_{\mathrm{j}},z,\kappa) \\ \mathrm{d}F_{\mathrm{yb,j}}(\xi_{\mathrm{j}},z,\kappa) \\ \mathrm{d}F_{\mathrm{zb,j}}(\xi_{\mathrm{j}},z,\kappa) \end{bmatrix} = A_1 \cdot A_2 \cdot \begin{bmatrix} \mathrm{d}F_{\mathrm{t,j}}(\xi_{\mathrm{j}},z,\kappa) \\ \mathrm{d}F_{\mathrm{r,j}}(\xi_{\mathrm{j}},z,\kappa) \\ \mathrm{d}F_{\mathrm{a,j}}(\xi_{\mathrm{j}},z,\kappa) \end{bmatrix} \quad (19)$$

where A_1 is the coordinate transformation matrix from normal plane to orthogonal plane. A_2 is the coordinate transformation matrix from orthogonal plane to tool center, which can be described as

$$A_1 = \begin{pmatrix} \cos i & 0 & \sin i \\ 0 & 1 & 0 \\ -\sin i & 0 & \cos i \end{pmatrix}^{-1} \quad (20)$$

$$A_2 = \begin{bmatrix} \cos\gamma & 0 & 0 \\ 0 & \cos\gamma & 0 \\ -\sin\gamma & -\sin\gamma & 1 \end{bmatrix} \begin{bmatrix} -\cos\xi_{\mathrm{j}} & -\sin\kappa\cdot sin\xi_{\mathrm{j}} & -\cos\kappa\cdot sin\xi_{\mathrm{j}} \\ \sin\xi_{\mathrm{j}} & -\sin\kappa\cdot cos\xi_{\mathrm{j}} & -\cos\kappa\cdot\cos\xi_{\mathrm{j}} \\ 0 & \cos\kappa & -\sin\kappa \end{bmatrix} \quad (21)$$

By integrating the axial height of the tool and the axial immersion angle, the cutting force on the j_{th} cutting edge can be obtained as follows

$$F_{\mathrm{ibj}} = \int_{z_{\min}}^{z_{\max}} \int_{\kappa_{\min}}^{\kappa_{\max}} \mathrm{d}F_{\mathrm{ib,j}}(\xi_{\mathrm{j}},\kappa,z)\mathrm{d}z\mathrm{d}\kappa, \quad i = x, y, z \quad (22)$$

Then coordinate system is transformed from tool coordinate system to workpiece center, Eq. (22) can be calculated as

$$\begin{bmatrix} F_{\mathrm{xaj}}(\xi_{\mathrm{j}}) \\ F_{\mathrm{ybj}}(\xi_{\mathrm{j}}) \\ F_{\mathrm{zbj}}(\xi_{\mathrm{j}}) \end{bmatrix} = A_3 \cdot \begin{bmatrix} F_{\mathrm{xbj}}(\xi_{\mathrm{j}}) \\ F_{\mathrm{ybj}}(\xi_{\mathrm{j}}) \\ F_{\mathrm{zbj}}(\xi_{\mathrm{j}}) \end{bmatrix} \quad (23)$$

$$A_3 = \begin{pmatrix} \cos\theta_{\mathrm{j}} & \sin\theta_{\mathrm{j}} & 0 \\ -\sin\theta_{\mathrm{j}} & \cos\theta_{\mathrm{j}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (24)$$

In the milling process of ball end cutter, when there is downward feed speed, indentation force will be generated [20]. According to material removal and chip volume, the indentation force can be calculated as by

$$F_{\mathrm{d}}(t) = K_{\mathrm{db}} \cdot \Delta V = K_{\mathrm{db}} \cdot \Delta S \cdot f_{\mathrm{t}} \quad (25)$$

where $F_{\mathrm{d}}(t)$ is the indentation force, ΔS is the cross-section area of the material to be cut, K_{db} is the cutting force coefficient related to indentation force, which can be obtained by linear fitting of the average axial force data measured in the helical milling process. In the workpiece coordinate system, the cutting force in the global coordinate system can be written as

$$
\begin{cases}
F_{\mathrm{X}}(\xi,\theta)=\sum_{j=1}^{N} g\left(\xi_{\mathrm{j}}\right)\left[F_{\mathrm{xaj}}\cos\theta_{\mathrm{j}}+F_{\mathrm{yaj}}\sin\theta_{\mathrm{j}}\right] \\
F_{\mathrm{Y}}(\xi,\theta)=\sum_{j=1}^{N} g\left(\xi_{\mathrm{j}}\right)\left[-F_{\mathrm{xaj}}\sin\theta_{\mathrm{j}}+F_{\mathrm{yaj}}\cos\theta_{\mathrm{j}}\right] \\
F_{\mathrm{Z}}(\xi,\theta)=\sum_{j=1}^{N} g\left(\xi_{\mathrm{j}}\right)\left(F_{\mathrm{aaj}}+F_{\mathrm{d}}(z)\right)
\end{cases}
\qquad
g\left(\xi_{\mathrm{j}}\right)=\begin{cases}0, & \xi_{\mathrm{en}}\le\xi_{\mathrm{j}}\le\xi_{\mathrm{ex}} \\ 1, & \text{otherwise}\end{cases}
\tag{26}
$$

where $g\left(\xi_{\mathrm{j}}\right)$ is the window function used to identify whether the micro-element is in cut or not.

4.5 Cutting force coefficient modeling

According to the oblique cutting theory, the inclination angle is equal to the helix angle of tool in ball helical milling, oblique cutting model of micro-element edge in ball helical milling is shown in Fig. 12.

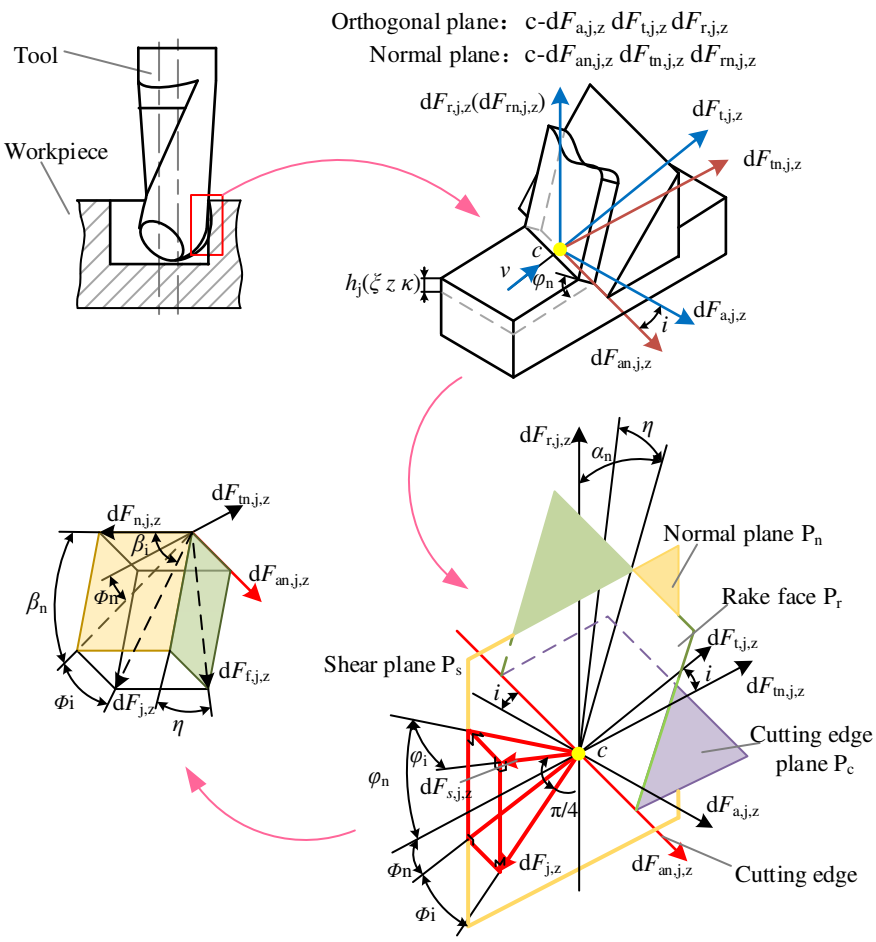

Fig. 12 Oblique cutting model of micro-element edge in ball helical milling

In the working normal plane reference system, the shear force coefficients proposed by Budak [13] are revised as follows

$$
\begin{cases}
K_{\mathrm{tc}}=\dfrac{\tau\left(\cos(\beta_{\mathrm{n}}-\alpha_{\mathrm{n}})+\tan\eta\sin\beta_{\mathrm{n}}\tan i\right)}{c\sin\varphi_{\mathrm{n}}} \\
K_{\mathrm{rc}}=\dfrac{\tau\left(\sin(\beta_{\mathrm{n}}-\alpha_{\mathrm{n}})\right)}{c\sin\varphi_{\mathrm{n}}\cos i} \\
K_{\mathrm{ac}}=\dfrac{\tau\left(\cos(\beta_{\mathrm{n}}-\alpha_{\mathrm{n}})\tan i-\tan\eta\sin\beta_{\mathrm{n}}\right)}{c\sin\varphi_{\mathrm{n}}} \\
c=\sqrt{\cos^{2}(\varphi_{\mathrm{n}}+\beta_{\mathrm{n}}-\alpha_{\mathrm{n}})+\tan^{2}\eta\sin^{2}\beta_{\mathrm{n}}}
\end{cases}
\tag{27}
$$

The cutting edge force coefficients K_{te}, K_{re}, K_{ae} can be estimated from the analytical model proposed by Abdelmoneim and Scrutton [21]

$$\begin{cases} K_{te} = r_e \tau (2\beta_n / \cos\beta_n + \pi \sin\beta_n \tan\beta_n) \\ K_{ae} = K_{te} \sin(i) \\ K_{re} = r_e \tau (2\sqrt{3} \sin\beta_n) \end{cases} \tag{28}$$

where the chip flow angle η was equal to the inclination angle i [22].

According to the principle of maximum shear stress, the included angle between the acting force and the shear plane should be $\pi/4$, as shown in Fig. 12, then

$$\begin{cases} \mathrm{d}F_{s,j,z} = \mathrm{d}F_{j,z}(\cos(\phi_n + \varphi_n)\cos\phi_i \cos\varphi_i + \sin\phi_i \sin\varphi_i) = \mathrm{d}F_{j,z} \cdot \cos\dfrac{\pi}{4} \\ \mathrm{d}F_{j,z}(\cos(\phi_n + \varphi_n)\cos\phi_i \sin\phi_i - \sin\phi_i \cos\varphi_i) = 0 \end{cases} \tag{29}$$

then

$$\sin\varphi_i = \sqrt{2}\sin\phi_i \tag{30}$$

$$\cos(\varphi_n + \phi_n) = \frac{\tan\phi_i}{\tan\varphi_i} \tag{31}$$

Shear stress τ can be further defined as

$$\tau = \mathrm{d}F_{s,j,z} / S_s \tag{32}$$

Where S_s is the shear area, which can be found that

$$S_s = f_t \cdot a_o / (\cos i \cdot \sin\varphi_n) \tag{33}$$

Besides, $\mathrm{d}F_{tn,j,z}$, $\mathrm{d}F_{rn,j,z}$ and $\mathrm{d}F_{an,j,z}$ are the normal tangential, radial and axial forces respectively, which can be obtained by

$$(\mathrm{d}F_{tn,j,z}, \mathrm{d}F_{rn,j,z}, \mathrm{d}F_{an,j,z})^{\mathrm{T}} = T(i)(\mathrm{d}F_{t,j,z}, \mathrm{d}F_{r,j,z}, \mathrm{d}F_{a,j,z})^{\mathrm{T}} \tag{34}$$

where $T(i)$ is a transformation matrix, which can be discribed as

$$T(i) = \begin{pmatrix} -\cos i & -\sin i & 0 \\ \sin i & -\cos i & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{35}$$

In addition,

$$\tan(\beta_n - \alpha_n) = \tan\phi_n = \mathrm{d}F_{rn,j,z} / \mathrm{d}F_{tn,j,z} \tag{36}$$

with

$$\alpha_n = \alpha \cdot \cos i \tag{37}$$

In the cutting process, the loads acting on the rake face of the tool include positive pressure $\mathrm{d}F_{n,j,z}$ and friction $\mathrm{d}F_{f,j,z}$, as shown in Fig. 12, then

$$\mathrm{d}F_{f,j,z} \sin\eta = \mathrm{d}F_{j,z} \sin\phi_i \tag{38}$$

$$\mathrm{d}F_{j,z} \cos\eta = \mathrm{d}F_{n,j,z} \tan(\phi_n + \gamma_n) \tag{39}$$

with

$$\sin\phi_i = \sin\beta_i \sin\eta \tag{40}$$

$$\tan(\phi_n + i) = \tan\beta_i \cos\eta \tag{41}$$

where

$$\phi_n + i = \beta_n \tag{42}$$

4.6 Calibration experiment of cutting force coefficients

When the rotating speed is 3183rpm, cutting depth is 0.2 mm/orb, and feed rate per tooth are 0.05, 0.06 and 0.08 mm/t respectively, three groups of helical milling experiments are selected to calibrate the cutting force coefficients.

Only a single edge participates in the cutting, axial depth of cut per revolution is 0.2 mm, the collected cutting force is equal to the cutting force on the micro-element edge [15]. It is transformed into the three-direction force on the micro-element edge by the inverse matrix, which can be calculated as

$$\begin{pmatrix} F_{\mathrm{T,j,a}}(\zeta) \\ F_{\mathrm{R,j,a}}(\zeta) \\ F_{\mathrm{A,j,a}}(\zeta) \end{pmatrix} = A_3^{-1} \cdot \begin{pmatrix} F_{\mathrm{X}}(\zeta) \\ F_{\mathrm{Y}}(\zeta) \\ F_{\mathrm{Z}}(\zeta) \end{pmatrix} \tag{43}$$

Eq. (43) is brought into the normal plane reference system of workpiece in Eq. (34), then Eqs (29)~(33) and (36)~(42) are solved together. The parameters β_{n} , ϕ_{n} , τ under the instantaneous undeformed chip thickness are received, thus the cutting force coefficients $K_{\mathrm{tc}}, K_{\mathrm{rc}}, K_{\mathrm{ac}}. K_{\mathrm{te}}, K_{\mathrm{re}}, K_{\mathrm{ae}}$ are obtained by taking it into Eq. (27) and (28), and they are expressed as the function of the instantaneous undeformed chip thickness by Weibull functions

$$K_{\mathrm{mn}}(\theta) = \exp\left[U - (U - V) \cdot \exp(-(a \cdot h(\theta))^{\mathrm{b}})\right] \tag{44}$$

where U, V, a, b are fit coefficients, which are illustrated in Table 2.

Table 2 Fitting coefficients of Weibull functions

	K_{tc}(N/mm^2)	K_{rc}(N/mm^2)	K_{ac}(N/mm^2)	K_{te}(N/mm)	K_{re}(N/mm)	K_{ae}(N/mm)
U	3.9760	2.4980	3.5750	0.7484	-1.252	0.1925
V	4.4580	4.1930	10.4200	2.3970	2.6930	1.8410
a	29.6200	12.9200	0.7164	12.0200	0.0008	12.0200
b	0.9297	0.9177	-1.117	0.9651	0.0646	0.9651

5 Model validation and analysis

The comparison diagrams of the filtered measured and simulated values of the full life-cycle cutting forces under three groups of different cutting conditions, as illustrated in Figs. 13 to 15, in which the duration and number of cycles of each stage are shown in Table 3.

Table 3 Duration and number of cycles for the three stages

	Stages	Duration(s)	Number of cycles
Fig.13	Cutting-in	65.67	27.78
	Steady	43.78	18.52
	Cutting-out	65.67	27.78
Fig.14	Cutting-in	24.93	16.76
	Steady	16.62	11.17
	Cutting-out	24.93	16.76
Fig.15	Cutting-in	23.29	16.76
	Steady	15.52	11.17
	Cutting-out	23.29	16.76

There are 27 local enlarged contrast maps, M in the figures denotes the measured value of cutting force, S means the simulated value.

When n_{rot}=3183 rpm, f_{t}=0.05 mm/t, a_{o}=0.1 mm/orb, the simulated cutting forces are compared with the measured values, the results are shown in Figure 13.

The machining time of single hole is 175.12 s, the cutting-in and cutting-out stages are both 65.67 s, and the steady stage is 43.78 s. Because the thickness of titanium alloy is 5 mm, the stable stage is shorter than the cutting-in stage, in other words, the machining depth of the steady stage is 2 mm, while the machining depth of the cutting-in and cutting-out stages are both equal to the tool radius of 3 mm.

At the cutting-in stage, the cutting forces of 6-11 s are selected for analysis. The maximum measured values of F_{X} and F_{Y} are 10 N, the maximum simulated values are 8.2 N, the maximum measured value of F_{Z} is 435.29 N, the maximum simulated value is 367.64 N, the errors of maximum values are 18%, 18% and 15.5%, respectively.

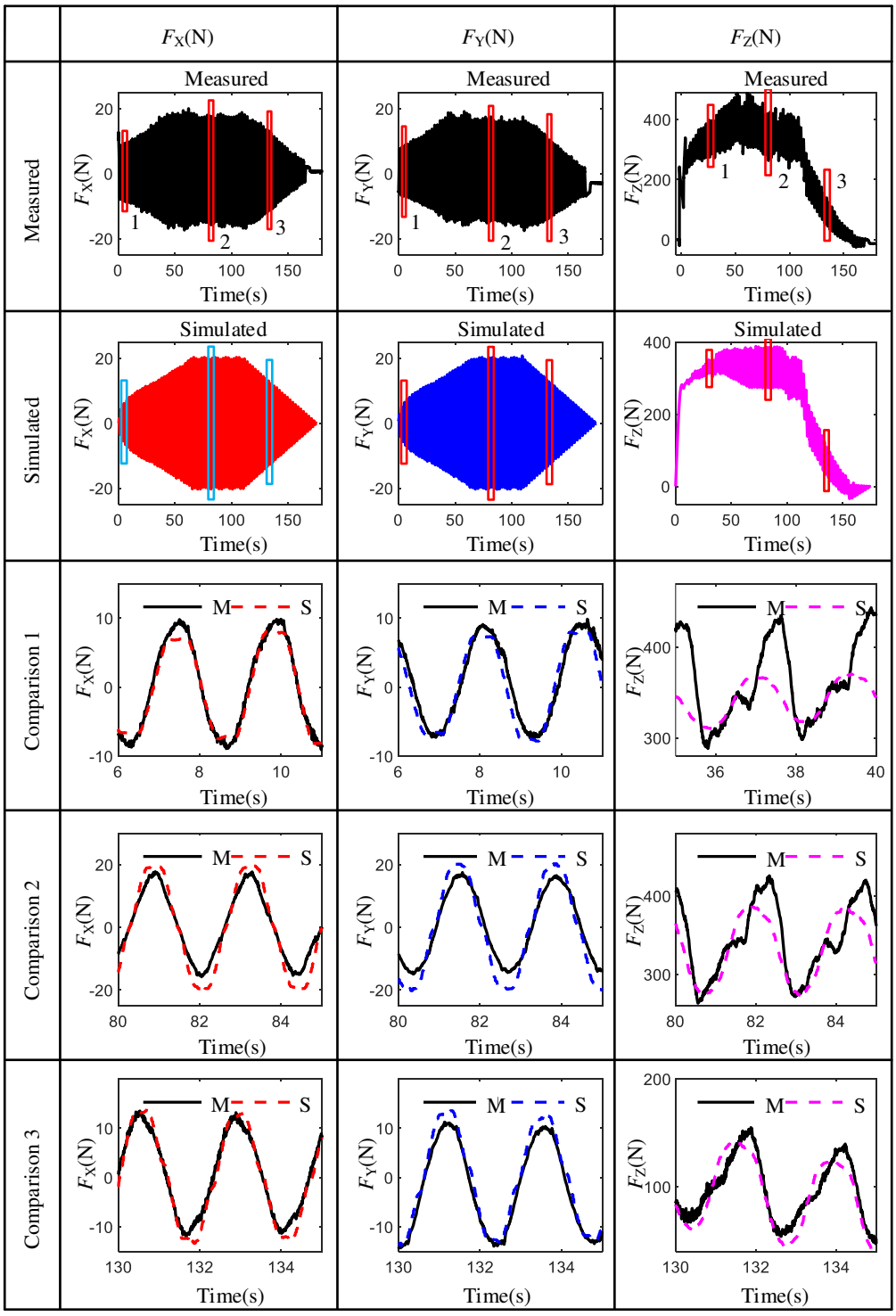

Fig.13 Comparisons of measured and simulated cutting forces

(n_{rot}=3183 rpm, f_t=0.05 mm/t, a_o=0.1 mm/orb)

During the steady stage, the cutting force from 80 to 85 s are analyzed. The maximum measured values of F_X and F_Y are both 17.39 N, the maximum simulated values are 20 N, the maximum measured value of F_Z is 425 N, the maximum simulated value is 389.29 N, the errors of maximum values are 15.01%, 15.01% and 8.4%, respectively.

In the cutting-out process, the forces comparison diagrams of 130-135 s are enlarged. The maximum measured values of F_X, F_Y and F_Z are 14.44 N, 14.44 N and 207.69 N respectively, and the maximum predicted values are 14.41 N, 14.41 N and 220.51 N severally. The errors of maximum values are 0.2%, 0.2% and 6.17%.

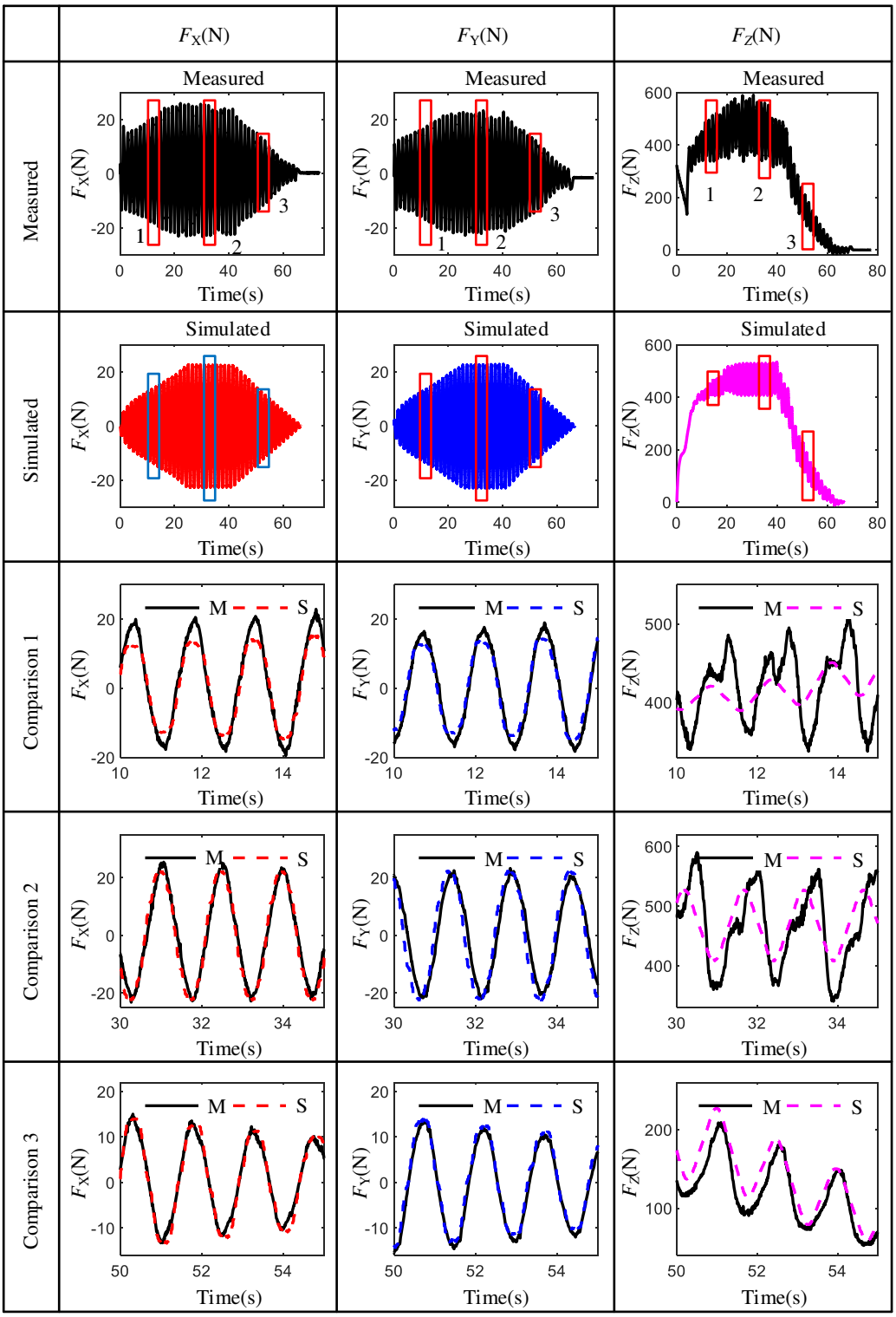

Fig.14 Comparisons of measured and simulated cutting forces

(n_{rot}=3183 rpm, f_t=0.08 mm/t, a_o=0.2 mm/orb)

It is noted that although there is a certain deviation between the predicted cutting forces and the measured values, the predicted values can capture the variation of cutting forces well and reflect the changes of cutting forces and direction in real time, the fluctuation period and amplitude of them are basically same, the F_X reaches peak before the F_Y.

It can be seen from the Figs. 13 to 15 that at the cutting-in stage, the measured cutting forces are greater than the predicted values in the first few orbital revolutions. The reasons maybe lie in when the cutter head is in contact with the workpiece, the linear

velocity of tip is very low, extrusion will occur, which will cause local deformation and plastic side flow of the workpiece material, so the F_X and F_Y are larger.

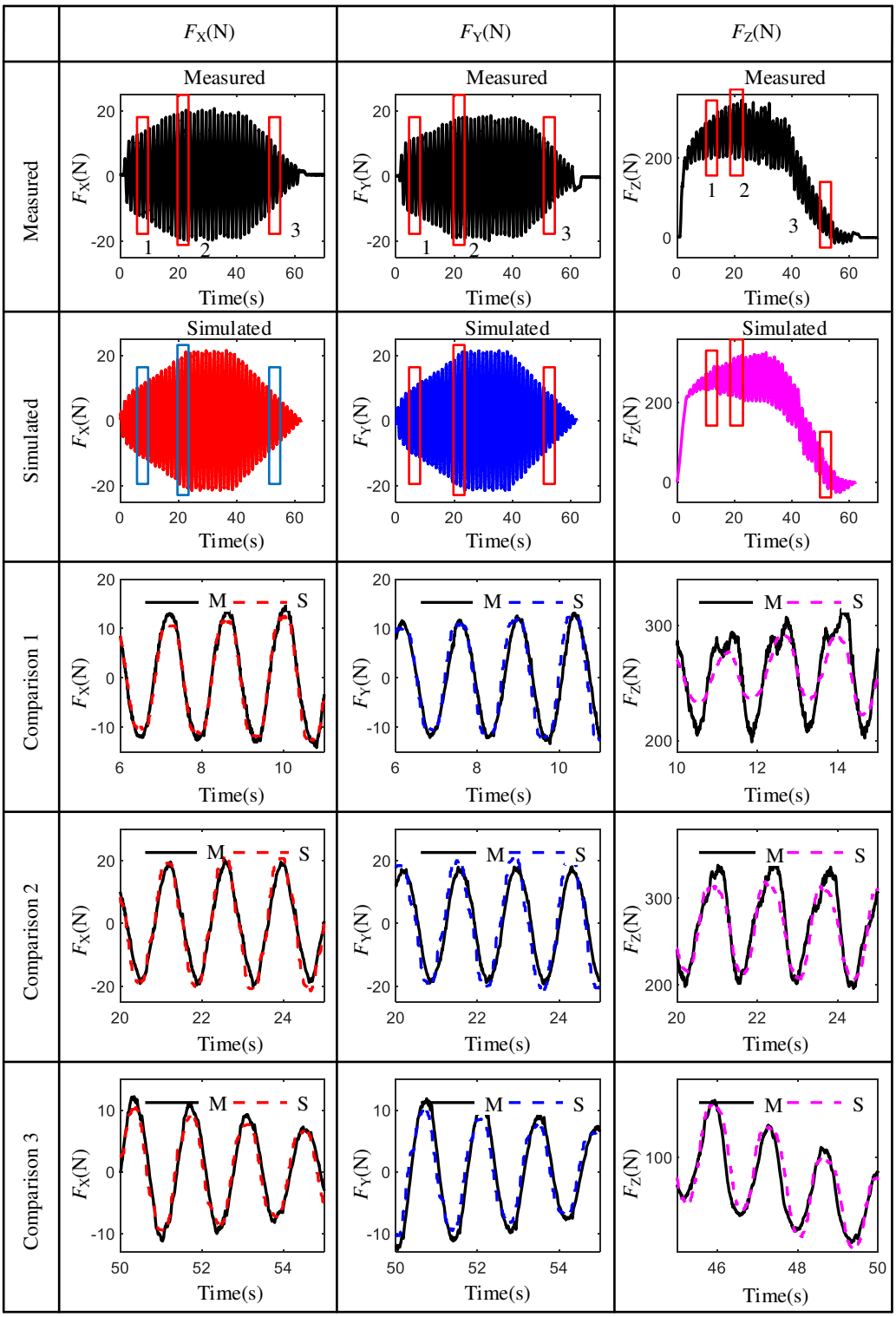

Fig. 15 Comparisons of measured and simulated cutting forces

(n_{rot}=4244 rpm, f_i=0.06 mm/t, a_o=0.2 mm/orb)

In the stable stage, the predicted cutting forces fit the measured value well, because the cutting depth is larger than the radius of ball end cutter, the amplitudes of three-direction forces reached the maximum, and the material removal volume remains stable, which is usually the focus of analysis. In addition, at the cutting-out stage, the predicted values fit well in the overall trend and size.

In order to further investigate the size of the deviation of cutting forces, the average error between the measured forces and the simulated values are calculated [23], the results are illustrated in Table 4.

Table 4 Errors of simulated and measured cutting forces

No.	E_x(%)	E_y(%)	E_z(%)
Fig.13	7.90	9.58	18.96
Fig.14	7.06	6.94	17.80
Fig.15	9.43	8.59	14.62
Average error	8.13	8.37	17.13

It can be seen that average errors of cutting forces in three directions are 8.13%, 8.37% and 17.13%, respectively, while the maximum errors are 9.43%, 9.58% and 18.96%, respectively. Namely, the maximum average error is less than 18%. While ignoring the influence of interference signals, the simulated values basically accurately fit the conventional change of measured cutting forces.

6 Conclusions

1. Depending on the motion vectors of helical milling and the geometric model of ball end cutter, the motion equation of any cutter location on the cutting edge is solved. The cutting process is divided into three stages, cutting-in, steady and cutting-out stages, the undeformed cutting area and volume in different stages are calculated.
2. In the machining process, considering that the fixed point on the cutting edge of the ball end cutter, the radial distance, the axial immersion angle and the radial position angle are not alike in different machining stages, that is, they change with the increase of cutting depth, the full life-cycle cutting force model of ball helical milling is constructed.
3. The cutting force coefficient model is constructed by utilizing the oblique cutting theory. Based on the geometric relationships and force analysis in oblique cutting process, the constraint equations of friction angle, shear angle and shear stress are derived. Through the calibration experiments of cutting force coefficient, six cutting force coefficients are calculated and expressed as Weibull functions of instantaneous undeformed chip thickness, the indentation cutting force coefficient K_{db} can be acquired by linear fitting of the average axial force.
4. The measured and predicted cutting forces fit well with the maximum average error less than 18%. The predicted cutting forces can be used to estimate the variation of cutting forces under different cutting parameters in the ball helical milling of titanium alloy.

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Availability of data and material:

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability: Not applicable

Author contributions: Not applicable

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent to publish Not applicable.

Competing interests The authors declare no competing interests.

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